

Groq Investment Memo

We're incredibly excited to launch this opportunity to invest into Groq's Series D, the company with potential to disrupt Nvidia. Groq has made headlines with its record-breaking LPU (Language Processing Unit), which it reports runs large language models (LLMs) over 4x faster, and at 1/5th the cost than using Nvidia GPUs - delivering over 20x better price/performance. This performance gap should widen with its next generation chip and as models get larger.

In the AI age, compute is the new oil. There will always be demand for compute at the fastest speed and lowest cost - exactly where Groq is positioned.

Highlights

- **Market-leading product:** High speed inference is essential to the exploding AI market (\$1.1T market by 2027). Groq is leading by a huge margin. Their LPU performed so well in [recent independent benchmarks](#) that the axes had to be extended to plot Groq on the Latency vs. Throughput chart.
- **Strong demand for real-time inference:** Once an LLM is trained, pretty much all use cases need real-time inference and inference speed is a major factor as companies bring LLMs to the masses. GPU-based inference is slow. GroqCloud, its usage based access to LPUs, solves this. Launched in March 2024, GroqCloud is already used by 170,000+ developers and more than 19,000 new apps.
- **Exponential growth potential:** Groq's CEO has stated that they are on track to deploy the equivalent of all of OpenAI's tokens per second in the next twelve months. Alongside its large developer ecosystem on GroqCloud, Groq has recently signed partnerships with sovereigns including Saudi Aramco and Norway's EWP (Earth Wind Power). **Groq could generate more revenue in 2025 than its current round valuation.**
- **Disrupting one of the largest and fastest growing markets:** the \$54B global AI hardware market already supports multiple \$1T+ & \$100B+ companies, and is set to grow 9x over the next decade to \$474B. Groq estimates that once an AI model/app reaches maturity they allocate 90-95% of resources to inference, giving Groq a huge total addressable market.
- **Industry-leading performance even on commodity silicon vs Nvidia's high-end hardware:** Groq's LPUs are built with commodity supply chain hardware as Nvidia bought out all high-end silicon component supply chains. Still, Groq chips demonstrated excellent performance - it'll only improve as supply chains get more relaxed. Groq is the only AI chip with its supply chain entirely in North America, both shielding it from geopolitical issues (e.g.: China - Taiwan conflict) and making it a preferred supplier for lucrative US government contracts.

- **Stellar team with proven track record:** Groq is led by founder and CEO Jonathan Ross, who invented the Tensor Processing Unit at Google. The team is laser focused on high speed, low cost inference - whilst it faces competition from Nvidia, Groq is readying its next generation chip and expects its advantages over Nvidia to widen as both companies release new chips.

Company

Groq is a generative AI solutions company and the creator of the LPU Inference Engine, the fastest language processing accelerator on the market. Groq was founded in 2016 by Jonathan Ross, who invented a new chip at Google called the Tensor Processing Unit (TPU) that was deployed across Google's data centers within a year. The TPU was a huge success, eventually underpinning more than 50% of all of Google's compute power.

Since its founding, Groq has raised \$362m from investors including Social Capital and in March 2024 announced its acquisition of Definitive Intelligence to form a new division - Groq Systems - focused on expanding its customer and developer ecosystem. In Groq's last funding round it was valued at \$1B.

Product

Groq chips are purpose-built to function as dedicated language processors. Large language models such as OpenAI's GPT-4 work by analyzing a sequence of words; then, using those words, they predict the next term in sequence. How accurate they are in predicting the next word is a critical factor for determining the best model.

Groq chips are optimized for the sequential nature of natural language and other sequential data like music and code. Being so specific in their design leads to much better performance on language tasks than, for example, GPUs that are optimized for parallel graphics processing. Groq estimates that it has a 4x to 100x speed advantage compared to other systems.

Whilst GPUs work well for batch inference in tasks like code analysis, document analysis and batch translations, LPUs are superior for a wider range of high impact use cases. AI agent workflows like software co-pilots, fraud prevention, immersive gaming and non-player characters (NPCs), and voice assistants need real-time inference, as do chatbots and virtual assistants.

The LPU's superior performance has been recognized in independent benchmarks, where it led in key performance indicators, further solidifying Groq's position as a leader in AI acceleration. The Groq LPU™ Inference Engine performed so well with a leading open source LLM from Meta AI, Llama 2-70b, that axes had to be extended to plot Groq on the Latency vs. Throughput chart.

Whilst Groq offers hardware sales, its GroqCloud offering abstracts away this complexity with a simple API to offer a vertically integrated AI inference solution.

Why now

Whilst Groq may seem like an overnight success, it was founded in 2016 and has years of R&D behind it, in both hardware and software. Groq's founder Jonathan Ross explained that the boom in generative AI has caused new problems to emerge that Groq is perfectly positioned to solve:

*"No one buys something because something is better, but because they have problems that are unsolved. You have to solve an unsolved problem. And until recently, we were going to people and giving them a problem saying we could lower your cost if you switch to our chips, or we could speed things up and they were telling us it's fast enough, it's cheap enough, you're just giving me a problem. But now, people have these models that they cannot run fast enough. And so we're solving their problem, and that's a very different sales motion. Up until we got a demo for large language model inference about two months ago, we had zero interest. Now, we are beating people away with a stick and we are having fights internally over how we allocate the hardware to customers. Our first 40 racks are already allocated, and **we are on track to deploy what we believe is the equivalent of all of OpenAI's tokens per second in the next twelve months** as our plan of record, and may deploy more than that. We have an unencumbered supply chain – we don't have HBM, we don't have CoWoS, so we're not competing with all of them for those technologies."*

Ross has stated that Groq can deploy 220,000 LPUs this year. Each LPU is on sale for \$20,000, although Groq is prioritizing its cloud service with a Tokens-as-a-Service (TaaS) offering. Since launching GroqCloud on 1 March 2024, over 170,000 new developers are using GroqCloud and more than 19,000 new applications are running on the LPU Inference Engine via the Groq API.

For large LPU orders, Groq is partnering with customers on data centre deployments. It has recently signed partnerships with sovereigns including Saudi Aramco and [Norway's EWP](#) (Earth Wind Power). The latter could see Groq deploy 130,000 LPUs by the end of 2025. Ross has stated that they intend to deploy more than 1.5 million LPUs world-wide by this time, with only 1 million unallocated at the moment - a number that is ["getting rapidly eaten up now that people know we're doing big deals."](#)

Market & competition

The global artificial intelligence hardware market size was estimated at \$54B in 2023 and is anticipated to reach around \$474B by 2033, expanding at CAGR of 24.5%. The AI chip market is dominated by three companies, Nvidia (\$2.6T), Intel (\$190B), and AMD (\$300B). They face competition from cloud platforms like AWS, Google Cloud, and Microsoft Azure, bringing proprietary AI chips in a box to market, and startups like Groq with specialized AI chips.

Groq has two key advantages over competitors. Firstly, all GPU manufacturers are subject to supply shortages of a component called HBM (High Memory Bandwidth), which are not used by LPUs. Alongside being insulated from HBM supply shortages, Groq's LPU is the only AI chip designed, engineered, and manufactured entirely in North America, shielding the company from

potential supply chain issues caused by geopolitical instability. More significantly, this should make Groq a preferred supplier for lucrative US government contracts.

Secondly, the GPU war is all about software: AMD GPUs are actually faster than Nvidia's - Nvidia's moat is not hardware, it's software, supported by 50,000+ CUDA kernel developers. LPU's were designed backwards from the software, which allows them to easily adapt to the rapid pace of innovation, while Nvidia GPUs require custom CUDA kernels, making bringing a new model to them more difficult.

Nvidia dominates the AI chip market today and recently unveiled its next generation Blackwell GPUs which it estimates to have up to 25 times less cost and energy consumption. However, Groq is readying its next generation chip that CEO Ross says will have up to 20 times improvement in power efficiency. Overall, Groq expects its advantage in inference to widen as both companies release new chips.

In addition to this, Groq is well-positioned for AI models getting larger - Anthropic CEO Dario Amodei says the cost of AI models will soon scale from \$100 million to the \$100 billion range - where it expects its speed and cost advantage will widen.

Further reading

GPUs vs LPUs:

<https://cryptoslate.com/groq-20000-lpu-card-breaks-ai-performance-records-to-rival-gpu-led-industry/>

<https://sacra.com/c/groq/>

<https://www.nextplatform.com/2023/11/27/groq-says-it-can-deploy-1-million-ai-inference-chips-in-two-years/>

<https://chamath.substack.com/p/the-future-of-ai-compute-a-conversation>

Benchmarks:

https://wow.groq.com/news_press/groq-lpu-inference-engine-leads-in-first-independent-llm-benchmark/

https://www.linkedin.com/posts/theomart_what-you-need-to-know-about-groq-and-lpus-activity-7172607328389685248-CtuH/

<https://www.prnewswire.com/news-releases/groq-acquires-definitive-intelligence-to-launch-groq-cloud-302077413.html>

<https://www.prnewswire.com/news-releases/demand-for-real-time-ai-inference-from-groq-accelerates-week-over-week-302105117.html>

Hardware vs cloud: <https://www.eetimes.com/groq-ceo-we-no-longer-sell-hardware/>

Risks & Disclosures

- *Groq is not yet a publicly listed company, making it difficult to buy and sell stock.*
- *Groq faces strong competition from other companies, including Nvidia.*
- *Groq has generated minimal revenue to date and its ability to scale revenue is not yet proven.*
- *The company faces significant technical and operational risks in both developing its product and manufacturing its product at scale.*
- *Investing in startups at this stage is incredibly risky - only invest what you're comfortable losing.*
- *The success of Groq projects is not guaranteed. Any setbacks or failures could negatively impact the company's valuation and result in complete loss of capital.*
- *As an indirect investment, we may have limited voting rights and no right to take part in the management of an Underlying Fund or ability to exercise control.*
- *The information in this memo is from publicly available sources and other investors may have been provided more due diligence materials.*
- *The company may be dependent on key members of leadership, e.g. the CEO, to achieve its goals, and should key members no longer work / be able to work at the company then it may impact business results.*